

Álgebra Relacional y SQL

Base de Datos

2 de julio de 2026

1. Sección 1: SQL $\rightarrow$ Álgebra Relacional

1. Colores de los botes reservados por Albert.

$$\pi_{color}(\sigma_{sname='Albert'}(sailors \bowtie reserves \bowtie boats))$$

2. sid de marineros con rating ≥ 8 o que reservaron el bote 103.

$$\pi_{sid}(\sigma_{rating \geq 8}(sailors)) \cup \pi_{sid}(\sigma_{bid=103}(reserves))$$

3. Nombres de marineros que no reservaron un bote rojo.

$$\pi_{sname}(sailors \bowtie (\pi_{sid}(sailors) - \pi_{sid}(\sigma_{color='red'}(boats \bowtie reserves))))$$

4. sid de marineros con edad ≥ 20 que no reservaron un bote rojo.

$$\pi_{sid}(\sigma_{age \geq 20}(sailors)) - \pi_{sid}(\sigma_{color='red'}(boats \bowtie reserves))$$

5. Nombres de marineros que reservaron al menos dos botes distintos.

$$\pi_{sname}(sailors \bowtie \pi_{sid}(\sigma_{r1.bid \neq r2.bid}(\rho_{r1}(reserves) \bowtie_{r1.sid=r2.sid} \rho_{r2}(reserves))))$$

6. Nombres de marineros que reservaron todos los botes.

$$\pi_{sname}(sailors \bowtie (\pi_{sid,bid}(reserves) \div \pi_{bid}(boats)))$$

7. Nombres de marineros que reservaron todos los botes llamados *BigBoat*.

$$\pi_{sname}(sailors \bowtie (\pi_{sid,bid}(reserves) \div \pi_{bid}(\sigma_{bname='BigBoat'}(boats))))$$

8. Nombres de proveedores que suministran alguna parte roja.

$$\pi_{sname}(\sigma_{color='red'}(suppliers \bowtie catalog \bowtie parts))$$

9. sid de proveedores que suministran alguna parte roja o verde.

$$\pi_{sid}(\sigma_{color='red' \vee color='green'}(catalog \bowtie parts))$$

10. sid de proveedores que suministran una parte roja o están en 21 George Street.

$$\pi_{sid}(\sigma_{color='red'}(catalog \bowtie parts)) \cup \pi_{sid}(\sigma_{address='21GeorgeStreet'}(suppliers))$$

11. sid de proveedores que suministran alguna parte roja y alguna verde.

$$\pi_{sid}(\sigma_{color='red'}(catalog \bowtie parts)) \cap \pi_{sid}(\sigma_{color='green'}(catalog \bowtie parts))$$

12. Pares de sid tales que el primero cobra más que el segundo por la misma parte.

$$\pi_{c1.sid, c2.sid}(\sigma_{c1.cost > c2.cost}(\rho_{c1}(catalog) \bowtie_{c1.pid=c2.pid} \rho_{c2}(catalog)))$$

13. sid de proveedores que suministran solo partes rojas.

$$\pi_{sid}(suppliers) - \pi_{sid}(\sigma_{color \neq 'red'}(catalog \bowtie parts))$$

14. sid de proveedores que suministran todas las partes.

$$\pi_{sid, pid}(catalog) \div \pi_{pid}(parts)$$

15. Nombres y salarios de jefes que tienen algún empleado con salario >100.

$$\pi_{b.name, b.salary}(\sigma_{e.salary > 100}(\rho_b(employees) \bowtie_{b.number=sv.boss} \rho_{sv}(supervises) \bowtie_{sv.employee=e.number} \rho_e(employees)))$$

16. Pares (jefe, empleado) donde el empleado gana más que su jefe.

$$\pi_{b.name, e.name}(\sigma_{e.salary > b.salary}(\rho_{sv}(supervises) \bowtie_{sv.boss=b.number} \rho_b(employees) \bowtie_{sv.employee=e.number} \rho_e(employees)))$$

17. Nombres de empleados que no tienen jefe.

$$\pi_{name}(employees \bowtie (\pi_{number}(employees) - \rho_{number}(\pi_{employee}(supervises))))$$

18. Hora de cita y nombre del cliente para las citas de *Giuliano* el 2026-02-14.

$$\pi_{c.name, a.atime}(\sigma_{a.adate='2026-02-14' \wedge s.name='Giuliano'}(\rho_a(appointments) \bowtie_{a.sid=s.sid} \rho_s(staff) \bowtie_{a.cid=c.cid} \rho_c(clients)))$$

19. Servicios que han sido solicitados al menos una vez.

$$\pi_{service}(appointments)$$

20. Clientes (nombre y teléfono) que nunca tomaron el servicio *manicure*.

$$\pi_{name, phone}(clients \bowtie (\pi_{cid}(clients) - \pi_{cid}(\sigma_{service='manicure'}(appointments))))$$

2. Sección 2: Álgebra Relacional → SQL

Ejercicio 1

```
SELECT sid
FROM sailors
WHERE rating > 7;
```

Ejercicio 2

```
SELECT sname
FROM sailors
WHERE age >= 18 AND age <= 25;
```

Ejercicio 3

```
SELECT DISTINCT s.sname
FROM sailors s
JOIN reserves r ON s.sid = r.sid
JOIN boats b ON r.bid = b.bid
WHERE b.color = 'red';
```

Ejercicio 4

```
SELECT DISTINCT s1.sid
FROM sailors s1
JOIN sailors s2 ON s1.rating > s2.rating
WHERE s2.sname = 'Bob';
```

Ejercicio 5

```
SELECT sid FROM sailors
EXCEPT
SELECT s1.sid
FROM sailors s1
JOIN sailors s2 ON s1.rating < s2.rating;
```

Ejercicio 6

```
SELECT pname
FROM parts
WHERE color = 'red';
```

Ejercicio 7

```
SELECT DISTINCT c.cost
FROM parts p
JOIN catalog c ON p.pid = c.pid
WHERE p.color = 'red' OR p.color = 'green';
```

Ejercicio 8

```
SELECT DISTINCT c.sid
FROM parts p
JOIN catalog c ON p.pid = c.pid
WHERE p.color IN ('red', 'green');
```

Ejercicio 9

```
SELECT DISTINCT s.sname
FROM parts p
JOIN catalog c ON p.pid = c.pid
JOIN suppliers s ON c.sid = s.sid
WHERE p.color IN ('red', 'green');
```

Ejercicio 10

```
SELECT DISTINCT s.sname
FROM parts p
JOIN catalog c ON p.pid = c.pid
JOIN suppliers s ON c.sid = s.sid
WHERE p.color = 'red' AND c.cost < 100;
```

Ejercicio 11

```
SELECT s.sname
FROM parts p
JOIN catalog c ON p.pid = c.pid
JOIN suppliers s ON c.sid = s.sid
WHERE p.color = 'red' AND c.cost < 100
INTERSECT
SELECT s.sname
FROM parts p
JOIN catalog c ON p.pid = c.pid
JOIN suppliers s ON c.sid = s.sid
WHERE p.color = 'green' AND c.cost < 100;
```

Ejercicio 12

```
SELECT c.sid
FROM parts p
JOIN catalog c ON p.pid = c.pid
WHERE p.color = 'red' AND c.cost < 100
INTERSECT
SELECT c.sid
FROM parts p
JOIN catalog c ON p.pid = c.pid
WHERE p.color = 'green' AND c.cost < 100;
```

Ejercicio 13

```
SELECT sid FROM suppliers
EXCEPT
SELECT sid FROM catalog;
```

Ejercicio 14

```
SELECT c.sid, p.pid
FROM (SELECT DISTINCT sid FROM catalog) c
CROSS JOIN (SELECT DISTINCT pid FROM parts) p
EXCEPT
SELECT sid, pid FROM catalog;
```

Ejercicio 15

```
SELECT name
FROM employees
WHERE salary > 1000 AND age < 30;
```

Ejercicio 16

```
SELECT DISTINCT b.name
FROM employees b
JOIN supervises sv ON b.number = sv.boss;
```

Ejercicio 17

```
SELECT number FROM employees
EXCEPT
SELECT boss FROM supervises;
```

Ejercicio 18

```
SELECT DISTINCT c.name, c.phone
FROM clients c
JOIN appointments a ON c.cid = a.cid
WHERE a.service = 'haircut';
```

Ejercicio 19

```
SELECT cid FROM clients
EXCEPT
SELECT cid FROM appointments;
```

Ejercicio 20

```
SELECT c.name, s.name  
FROM clients c  
JOIN appointments a ON c.cid = a.cid  
JOIN staff s ON a.sid = s.sid;
```